FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

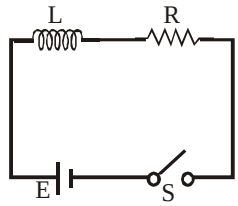
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two capillary tubes of radii 0.2 cm and 0.4 cm are dipped in the same liquid. The ratio of heights through which liquid will rise in the tube is
(A) 1:2 (B) 2:1
(C) 1:4 (D) 4:1

2. A particle executes S.H.M. with a time period of 4 s. Find the time taken by the particle to go directly from its mean position to half of its amplitude.
(A) 1 s (B) $\frac{1}{2}$s
(C) $\frac{1}{3}$s (D) $\frac{1}{4}$s

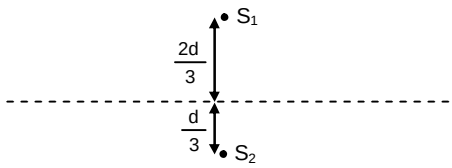
3. In the LR circuit as shown in figure, the switch is closed at t = 0. Mark the incorrect statement of the following:
(A)The current increases with time
(B)The current increases with a decreasing rate
(C)The voltage drop across the inductor increases
(D)The voltage drop across the resistor increases



4. In hydrogen like atoms the ratio of difference of energies $E_{4n} - E_{2n}$ and $E_{2n} - E_n$ varies with atomic number z and principle quantum number n as
(A) $\frac{z^2}{n^2}$ (B) $\frac{z^4}{n^4}$
(C) $\frac{z}{n}$ (D) none of these

Space for Rough work

5. Two coherent point sources of frequency ($f = \frac{10v}{d}$ where v is speed of light) are placed at a distance d apart as shown in figure. The receiver is free to move along the dotted line shown in the figure. Find total number of maxima observed by receiver.



(A) 6 (B) 7
(C) 5 (D) 8

6. The temperature of an ideal gas is increased for 100k to 400k. If at 100K the root mean square velocity of the gas molecules is v, at 400K it becomes
(A) 4v (B) 2v
(C) v/2 (D) v/4

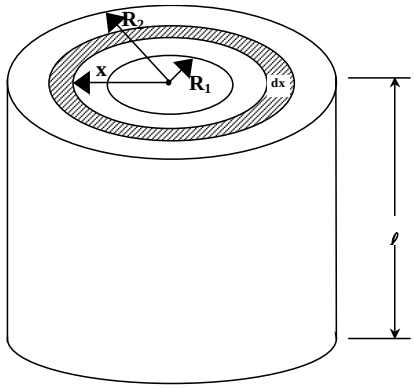
7. A boat which has a speed of 6 km/h in still water crosses a river of width 1 km along the shortest possible path in 20 min. The velocity of the river water in km/h is
(A) 1 (B) 3
(C) 4 (D) $3\sqrt{3}$

8. If photons of energy 12.75 eV are passing through hydrogen gas in ground state then no. of lines in emission spectrum will be
(A) 6 (B) 4
(C) 3 (D) 2

9. The momentum of α-particles moving in a circular path of radius 10 cm in a perpendicular magnetic field of 0.05 Tesla will be:
(A) 1.6×10^{-20} kg m/s (B) 1.6×10^{-21} kg m/s
(C) 1.6×10^{-19} kg m/s (D) 1.6×10^{-18} kg m/s

Space for Rough work

10. A cylindrical conductor of length ℓ and inner radius R_1 and outer radius R_2 has specific resistance ρ. A cell of emf ε is connected across the two lateral faces of the conductor. Find the current drown from the cell.



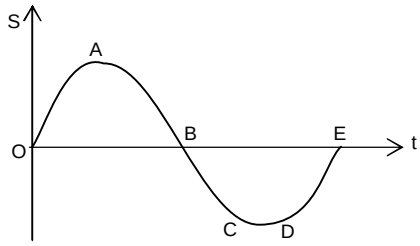
(A) $\dfrac{2\pi\ell\varepsilon}{\rho \ln \dfrac{R_2}{R_1}}$

(B) $\dfrac{2\pi\ell\varepsilon}{\rho (R_2^2 - R_1^2)}$

(C) $\dfrac{2\ell\varepsilon}{\rho\pi(R_2^2 - R_1^2)}$

(D) None of thee

11. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements is incorrect with respect to the motion?



(A) in motion between 0 to A, the velocity is positive and acceleration is negative

(B) between A and B, the velocity and acceleration are positive

(C) between B and C, the velocity is negative and acceleration is positive

(D) between D and E, the acceleration is positive

12. Force acting on a particle of mass m moving in straight line varies with the velocity of the particle as F = K/V here K is constant then speed of the particle in time t.

(A) $\sqrt{\dfrac{kt}{m}}$ (B) $\sqrt{\dfrac{2kt}{m}}$

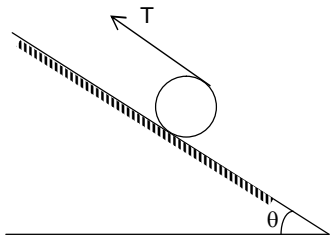
(C) $2\sqrt{\dfrac{kt}{m}}$ (D) None of these

Space for Rough work

13. The current in a wire varies with time according to the equation I = 4 + 2t, where I is in ampere and t is in sec. The quantity of charge which has passed through a cross-section of the wire during the time t = 2 sec to t = 6 sec will be
(A) 60 coulomb (B) 24 coulomb
(C) 48 coulomb (D) 30 coulomb

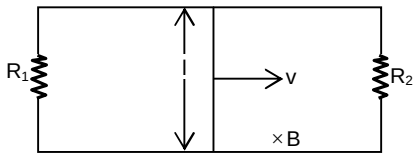
14. If ρ is the density of the material of a wire and σ is the breaking stress. The greatest length of the wire that can hang freely without breaking is
(A) $\frac{2\sigma}{\rho g}$ (B) $\frac{\rho}{\sigma g}$
(C) $\frac{\rho g}{2\sigma}$ (D) $\frac{\sigma}{\rho g}$

15. A sphere kept on a rough inclined plane is in equilibrium by a string wrapped over it. If angle of inclination is θ, the tension in string will be equal to



(A) mg sinθ (B) 2 mg sinθ
(C) $\frac{mg\sin\theta}{2}$ (D) mg cosθ

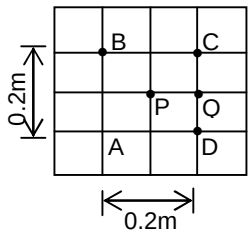
16. A rectangular loop with a slide wire of length l is kept in a uniform magnetic field as shown in figure. The resistance of slider is R. neglecting self inductance of the loop find the current in the connector during its motion with a velocity v.



(A) $\frac{Blv}{R_1+R_2+R}$ (B) $\frac{Blv(R_1+R_2)}{R+(R_1+R_2)}$
(C) $\frac{Blv(R_1+R_2)}{RR_1+RR_2+R_1R_2}$ (D) $Blv\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$

Space for Rough work

17. A, B, C, D, P and Q are points in a uniform electric field. The potentials at these points are V(A) = 2 volt. V(P) = V(B) = V(D) = 5 volt. V (C) = 8 volt. The electric field at P is



(A) $10 Vm^{-1}$ along PQ (B) $5 Vm^{-1}$ along PC

(C) $15\sqrt{2}$ Vm^{-1} along PA (D) 5 Vm^{-1} along PA

18. A concave mirror of focal length 15 cm is placed in water of refractive index μ = 4/3. Water is filled upto 30 cm. An object of height 1 cm is placed at a distance of 20 cm in front of the mirror. If the size of image is n times size of object, then n =

(A) 3 (B) 2

(C) 2.5 (D) 5

19. An ideal gas expands according to the law $PV^{3/2}$ =constant. We conclude that

(A) the adiabatic exponent of the gas k = 1.5

(B) the molar heat capacity $C = C_V - 2R$

(C) temperature increases during the process

(D) such a gas is not feasible

20. A tank has a hole made at its bottom. The time needed to empty the tank from level h_1 to h_2 will be proportional to

(A) $h_1 + h_2$ (B) $\sqrt{h_1} + \sqrt{h_2}$

(C) $h_1 - h_2$ (D) $\sqrt{h_1} - \sqrt{h_2}$

21. A source of sound emitting a note of frequency 200Hz moves towards an observer with a velocity v equal to the velocity of sound. If the observer also moves away from the source with the same velocity v, the apparent frequency heard by the observer is

(A) 50Hz (B) 100Hz

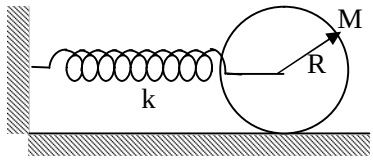
(C) 150Hz (D) 200Hz

Space for Rough work

22. A lake is covered with ice 2 cm thick. The temperature of ambient air is -15^oC. Find the rate of thickening of ice. For ice k = 4×10^{-4}k-cal-$m^{-1}s^{-1}(°C)^{-1}$. Density =0.9×10^3 kg/$m^3$and latent heat L = 80 Kilo Cal/Kg.
(A) 1.5 cm/h (B) 0.75 cm/h
(C) 3 cm/h (D) 2 cm/h

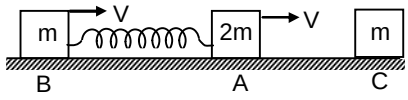
23. A cylindrical hall has a horizontal smooth floor. A ball is projected along the floor from A point on the wall in a direction making an angle θ with the radius through the point the ball returns back to the initial point after two impacts with the wall. If the coefficient of restitution is e then $\tan^2\theta$ will be
(A) $\frac{1+e+e^2}{e^3}$ (B) $\frac{1+e}{e^2}$
(C) $\frac{e^2}{1+e}$ (D) $\frac{e^3}{1+e+e^2}$

24. A solid cylinder of mass M is attached to the spring by means of a frictionless pivot. The cylinder is placed on floor. Find the time period of small oscillation of the system, if the floor is sufficiently rough to prevent any slipping of cylinder. Assume stiffness of spring to be K.



(A) $2\pi\sqrt{\frac{7M}{5K}}$ (B) $2\pi\sqrt{\frac{5M}{3K}}$
(C) $2\pi\sqrt{\frac{3M}{2K}}$ (D) $2\pi\sqrt{\frac{M}{K}}$

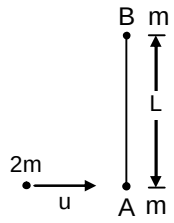
25. Two blocks A and B of mass 2m and m respectively are connected to a massless spring of spring constant K. If A and B moving on the horizontal frictionless surface with velocity v to right. If A collides with C of mass m elastically and head on, then the maximum compressions of the spring will be



(A) $\sqrt{\frac{3}{2}\frac{m}{k}}\,V$ (B) $\sqrt{\frac{27\,m}{8\,K}}\,V$
(C) $\sqrt{\frac{9\,m}{8\,K}}\,V$ (D) $\sqrt{\frac{8\,m}{27\,K}}\,V$

Space for Rough work

26. Two small balls A and B, each of mass m, are joined rigidly at the ends of a light rod of length L. They are placed on a frictionless horizontal surface. Another ball of mass 2 m moving with speed u towards one of the ball and perpendicular to the length of the rod on the horizontal frictionless surface as shown in the figure. If the coefficient of restitution is 1/2 then the angular speed of the rod after the collision will be



(A) $\frac{4}{3}\frac{u}{L}$ (B) $\frac{u}{L}$

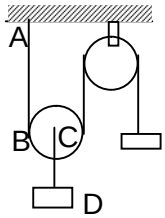
(C) $\frac{2}{3}\frac{u}{L}$ (D) None of these

27. A rod of negligible heat capacity has a length of 50 cm, area of cross-section 5 cm^2 and thermal conductivity 500 W/m^oC. The temperature of one end is maintained at 0°C and that of the other end is slowly and linearly varied from 0°C to 60°C in 20 min. Assuming no heat loss through the lateral side, the total heat transmitted through the rod in 20 minutes is

(A) 18 kJ (B) 9 kJ

(C) 27 kJ (D) 4.5 kJ

28. Both the strings shown in figure are made of same material and have same cross – section. The pulleys are light. The wave speed of a transverse wave in the string AB is v_1 and in CD it is v_2. Then v_1/v_2 is



(A) 1 (B) 2

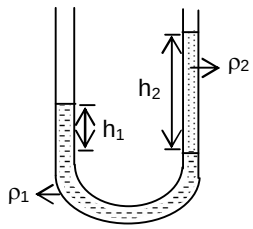
(C) $\sqrt{2}$ (D) $1/\sqrt{2}$

29. The excess pressure inside one soap bubble is four times that of other. Then the ratio of volume of the first bubble to other is

(A) 1 : 64 (B) 1 : 4

(C) 64 : 1 (D) 1 : 2

30. Two liquids which do not react chemically are placed in a bent tube as shown in figure. The heights of the liquids above their surface of separation are



(A) directly proportional to their densities

(B) inversely proportional to their densities

(C) directly proportional to square of their densities

(D) inversely proportional to square of their densities

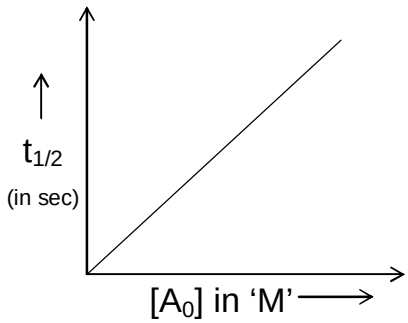
Space for Rough work

Chemistry PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The correct statement regarding the third orbit of hydrogen atom is
(A) it can accommodate a maximum of nine electrons
(B) it contains nine degenerate atomic orbitals
(C) it contains nine atomic orbitals having three sets of degeneracy
(D) it contains nine sub-shells

32. 

What is the unit of rate constant of the reaction, for which the above graph is given?
(A) s^{-1} (B) mol $L^{-1}s^{-1}$
(C) mol $L^{-1}s$ (D) mol^{-1} L^2s^{-1}

33. $SO_3(g) + CO(g) \rightleftharpoons SO_2(g) + CO_2(g)$
Which of the following activity can reduce the equilibrium concentration of CO to half?
(A) Increasing pressure twice
(B) Increasing the volume of reaction container to two times of its initial value
(C) Adding two moles of CO_2 gas
(D) Adding two moles of helium gas at constant pressure

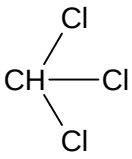
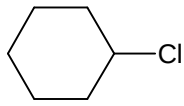
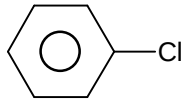
Space for Rough work

34. 200 mL of 0.4 M solution of CH_3COONa is mixed with 400 mL of 0.2 M solution of CH_3COOH. After complete mixing, 400 mL of 0.1 M NaCl is added to it. What is the pH of the resulting solution? [K_a of $CH_3COOH = 10^{-5}$]
(A) 5.4 (B) 6
(C) 5 (D) 6.2

35. Which of the following bond angle is not observed in PCl_5 molecule in gaseous state?
(A) 120° (B) 180°
(C) 90° (D) 60°

36. Which of the following produces a coloured gas on heating?
(A) $NaNO_3$ (B) $LiNO_3$
(C) $MgCO_3$ (D) Li_3N

37. Which of the following type of compound is mixed with detergent powder to provide brightness to fabrics?
(A) Borates (B) Peroxoborates
(C) Boranes (D) Borides

38. In which of the following compound, chlorine exerts the maximum number of electronic effects?
(A) 
(B) 
(C) $CH_2 = CH - CH_2Cl$
(D) 

39. The soda extract of which of the following compound gives the test of nitrogen?
(A) NH_2NH_2 (B) HNO_2
(C) $HCONH_2$ (D) All are correct

Space for Rough work

40. Which of the following linkage is present in "Teflon"?
(A) H – C (B) F – F
(C) C – C (D) C = C

41. Which of the following reaction does not produce H_3PO_4?
(A) Reaction between P_4O_{10} and H_2O
(B) Heating of H_3PO_3
(C) Heating of H_3PO_2
(D) Reaction between Ca_3P_2 and H_2O

42. Which of the following reaction produce Cl_2 gas?
(A) $NaCl + H_2SO_4 \longrightarrow$ (B) $KClO_3 \xrightarrow{\text{Heat}}$
(C) $HCl + O_3 \longrightarrow$ (D) $KCl + AgNO_3 \longrightarrow$

43. Which of the following is a colourless complex ion?
(A) $[Cu(NH_3)_4]^{2+}$ (B) $[Cr(H_2O)_6]^{3+}$
(C) $[FeF_6]^{3-}$ (D) $[Mn(H_2O)_6]^{3+}$

44. Which of the following metallurgical process is not needed in order to extract Zn from $ZnCO_3$?
(A) Calcination (B) Roasting
(C) Carbon reduction (D) Distillation

45. Which of the following salt can increase the boiling point of water by maximum extent?
(Assume equimolar quantities are added)
(Assume complete dissociation of the salts)
(A) NaCl (B) $CaCl_2$
(C) $AlCl_3$ (D) KCl

46. Which of the following ion has the highest coagulation power for the sol. which is obtained by the following reaction?
$FeCl_3\,(\text{excess}) + NaOH \longrightarrow Sol$
(A) Al^{3+} (B) SO_4^{2-}
(C) I^- (D) Na^+

Space for Rough work

47. A hydrogen electrode is prepared by using a sample of HCl solution with pH = 4 and H_2 gas at 1 atm pressure. What is the electrode potential of the electrode?
(A) 0.814 V (B) 0.236 V
(C) -0.06 V (D) -0.618 V

48. The heat of atomization of a compound XY_3 in gaseous state is E kJ mol^{-1}. What is the X – Y bond energy of kJ mol^{-1} unit?
(A) E (B) 3E
(C) $\frac{E}{3}$ (D) $\frac{E}{6}$

49. $P(g) + 2Q(g) \longrightarrow PQ_2(g);\ \Delta H = 18\,kJ\ mol^{-1}$
The entropy change of the above reaction(ΔS_{system}) is 60 $JK^{-1}mol^{-1}$. At what temperature, the reaction becomes spontaneous?
(A) Below 200 K (B) Above 300 K
(C) Below 300 K (D) Above 200 K and below 300 K

50. 1.2 L of 0.4 M HCl solution can be completely neutralized by
(A) 600 mL of 1.2 M NaOH solution (B) 800 mL of 0.2 M NaOH solution
(C) 19.2 g of solid NaOH (D) 4.8 mole of NaOH

51. Which of the following should have the highest value for easily liquefiable gases?
(A) ab (B) $\frac{a}{b}$
(C) a + b (D) $\left(\frac{1}{a} + b\right)$

52. $(X) \xrightarrow{Ba^{2+}} \underset{\text{White ppt}}{Y\downarrow} \xrightarrow{Cl_2/H_2O} \underset{\text{White ppt}}{Z\downarrow}$
If the precipitate (Z↓) is $BaSO_4$, which of the following can be (X)?
(A) Na_2SO_4 (B) Na_2SO_3
(C) $NaHSO_4$ (D) Na_2CO_3

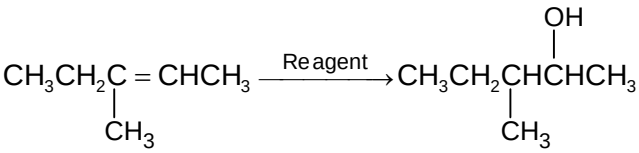
Space for Rough work

53. In the crystal lattice of CsCl, Cs^+ ions are present at the
(A) tetrahedral voids of the unit cell (B) octahedral voids of the unit cell
(C) body-centre of the unit cell (D) face-centre of the unit cell

54. Which of the following pollutant can form complex with haemoglobin of blood?
(A) CF_2Cl_2 (B) CO
(C) SO_3 (D) $CHCl_3$

55. Which of the following is an artificial sweetner?
(A) Sucrose (B) Sucrolose
(C) Frcutose (D) Maltose

56. Which of the following reagent can convert phenol(PhOH) to anisole($PhOCH_3$)?
(A) $CHCl_3$ (B) CH_2N_2
(C) CH_3Cl (D) HCHO

57. 

Which reagent forms the above organic product as the single product in the above reaction?

(A) H_2O/H^+ (B) $\frac{Hg(OCOCH_3)_2, H_2O}{NaBH_4}$

(C) $\frac{B_2H_6, THF}{NaOH, H_2O_2}$ (D) $KMnO_4/OH^-$/Cold

58. Which of the following compound forms two stereoisomers when treated with NH_2OH in weakly acidic medium?
(A) CH_3COCH_3 (B) HCHO
(C) $CH_3COC_2H_5$ (D) $C_2H_5COCH_2CH_3$

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59. $CH_3CH_2CH_2CONH_2 \xrightarrow{Br_2/KOH} Product$

The organic product of above reaction is

(A) $CH_3CH_2CH_2NH_2$

(B) $CH_3\underset{\displaystyle NH_2}{\underset{|}{C}}HCH_3$

(C) $CH_3CH_2CH_2CH_2NH_2$

(D) $CH_3CH_2\underset{\displaystyle NH_2}{\underset{|}{C}}HCH_3$

60. Which of the following compound has the highest boiling point?

(A) $CH_3CH_2COOCH_3$

(B) $CH_3CH_2CH_2CONH_2$

(C) $CH_3COOCOCH_3$

(D) CH_3CH_2COCl

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Mathematics PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $S = \frac{8}{5} + \frac{16}{65} + \frac{24}{325} + ... + \frac{128}{2^{18}+1}$, then

(A) $S = \frac{1088}{545}$ (B) $S = \frac{545}{1088}$

(C) $S = \frac{1056}{545}$ (D) $S = \frac{545}{1056}$

62. A function y = f(x) satisfies $(x+1).f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$. If f(0) = 5, then f(x) is

(A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$

(C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

63. If $\vec{a}$ and $\vec{b}$ are not perpendicular to each other and $\vec{r}\times\vec{a} = \vec{b}\times\vec{a}.\vec{r}.\vec{c} = 0$, then $\vec{r}$ is equal to

(A) $\vec{a} - \vec{c}$ (B) $\vec{b} + x\vec{a}$ for all scalars x

(C) $\vec{b} - \frac{(\vec{b}.\vec{c})}{(\vec{a}.\vec{c})}\vec{a}$ (D) None of these

Space for Rough work

64. If $f : R \to R$, $f(f(x)) = (f(x))^2$ then ffff(x) is equal to
(A) $(f(x^4))^2$ (B) $(f(x))^4$
(C) $(f(x^2))^2$ (D) $f(x^4)$

65. If α is a root of $x^2 + ax + 1 = 0$, then $\lim_{x \to 1/\alpha} \frac{\sin\left(x^2 + ax + 1\right)}{(\alpha x - 1)}$ is equal to
(A) $2a\alpha$ (B) $a\alpha^2$
(C) $\frac{1-\alpha^2}{\alpha^2}$ (D) does not exist

66. The differential coefficient of f(cosx) with respect to g (sinx) at $x = \pi/3$ if $f'(1/2)$, $g'(\sqrt{3}/2) = -2$ is equal to
(A) $2\sqrt{3}$ (B) $1/\sqrt{3}$
(C) $\sqrt{3}$ (D) $-\sqrt{3}$

67. If $f(x) = |1 - x|$, then the points where $\sin^{-1}(f|x|)$ is non–differentiable are
(A) {0, 1} (B) {0, – 1}
(C) {0, 1, – 1} (D) none of these

68. If the curve $y = x^2 + bx + c$ touches the straight line $y = x$ at the point (1, 1), then b and c are given by
(A) –1, 1 (B) –1, 2
(C) 2, 1 (D) 1, 1

69. $\int e^x \left(\frac{2\tan x}{1+\tan x} + \cot^2\left(x + \frac{\pi}{4}\right)\right) dx$ is equal to
(A) $e^x \tan\left(\frac{\pi}{4} - x\right) + c$ (B) $e^x \tan\left(x - \frac{\pi}{4}\right) + c$
(C) $e^x \tan\left(\frac{3\pi}{4} - x\right) + c$ (D) none of these

Space for Rough work

70. If $\int \frac{(\sqrt{x})^5}{(\sqrt{x})^7 + x^6} dx = a \ln \frac{x^k}{x^k + 1} + c$, the value of a and k respectively are

(A) $\frac{5}{2}$ and $\frac{2}{5}$ (B) $\frac{2}{5}$ and $\frac{5}{2}$

(C) $\frac{5}{2}$ and 2 (D) none of these

71. The value of $\int_0^{\pi/4} \ln \cos\left(\frac{\pi}{4} + x\right)^{\cot\left(\frac{\pi}{4} - x\right)} dx$ is

(A) 0 (B) 1

(C) 2 (D) not defined

72. The value of $\int_{-10}^{10} \frac{3^x}{3^{[x]}} dx$ is equal to

(A) 20 (B) $\frac{40}{\ln 3}$

(C) $\frac{20}{\ln 3}$ (D) none of these

73. If the intercept made on the line y = mx by the lines x = 2 and x =5 is less then 5, then the range of m is

(A) $\left(-\frac{4}{3}, \frac{4}{3}\right)$ (B) $\left(-\infty, -\frac{4}{3}\right) \cup \left(\frac{4}{3}, \infty\right)$

(C) $\left[-\frac{4}{3}, \frac{4}{3}\right)$ (D) none of these.

74. A pair of unbiased dice is rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes before 7 is

(A) 2/5 (B) 3/5

(C) 4/5 (D) none of these

Space for Rough work

75. The value of $\sum_{r=1}^{n+1}\left(\sum_{k=1}^{n} {}^{k}C_{r-1}\right)$ (where r, k, n $\in$ N) is equal to

(A) $2^{n+1}-2$ (B) $2^{n+1}-1$

(C) 2^{n+1} (D) none of these

76. If $b_{n+1}=\frac{1}{1-b_n}$ for $n \geq 1$ and $b_1 = b_3$, then $\sum_{r=1}^{2001} b_r{}^{2001}$ is equal to

(A) 2001 (B) –2001

(C) 0 (D) none of these

77. The ratio in which the plane $\vec{r}.\left(\vec{i}-2\vec{j}+3\vec{k}\right)=17$ divides the line joining the points $-2\vec{i}+4\vec{j}+7\vec{k}$ and $3\vec{i}-5\vec{j}+8\vec{k}$

(A) 1 : 5 (B) 1 : 10

(C) 3 : 5 (D) 3 : 10

78. A tetrahedron has vertices O (0, 0, 0), A (1, 2, 1), B (2, 1, 3) and C (–1, 1, 2), then angle between faces OAB and ABC will be:

(A) $\cos^{-1}\left(\frac{17}{31}\right)$ (B) 30°

(C) 90° (D) $\cos^{-1}\left(\frac{19}{35}\right)$

79. Which of the following are equations of the plane passing through the points P (1, 1, –1), Q (3, 0, 2) and R (–2, 1, 0)?

(A) $\left(2\hat{i}-3\hat{j}+3\hat{k}\right), \left(\left(x+2\right)\hat{i}+\left(y-1\right)\hat{j}+z\hat{k}\right)=0$

(B) $x=3-t,\ y=-11t,\ z=2-3t$

(C) $\left(x+2\right)+11\left(y-1\right)=3z$

(D) $\left(2\hat{i}-\hat{j}+3\hat{k}\right)\times\left(-3\hat{i}+\hat{k}\right).\left(\left(x+2\right)\hat{i}+\left(y-1\right)\hat{j}+z\hat{k}\right)=0$

Space for Rough work

80. The number of pairs of integer (x, y) that satisfy the following two equations $\begin{cases}\cos(xy)=x\\ \tan(xy)=y\end{cases}$ is

(A) 1 (B) 2
(C) 4 (D) 6

81. The total number of solutions of $\ln|\sin x| = -x^2 + 2x$ in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ is equal to

(A) 1 (B) 2
(C) 4 (D) none of these

82. Let AD be a median of the ΔABC. If AE and AF are medians of the triangle ABD and ADC, respectively. And $AD = m_1$, $AE = m_2$, $AF = m_3$, then $a^2/8$ is equal to

(A) $m_2^2 + m_3^2 - 2m_1^2$ (B) $m_1^2 + m_2^2 - 2m_3^2$
(C) $m_1^2 + m_3^2 - 2m_2^2$ (D) none of these

83. If A is order 3 square matrix such that $|A| = 2$, then $|\text{adj}(\text{adj}(\text{adj}\,A))|$ is

(A) 512 (B) 256
(C) 64 (D) none of these

84. Let $\{D_1, D_2, D_3, \ldots\ldots D_n\}$ be the set of third order determinants that can be made with the distinct non – zero real numbers $a_1, a_2 \ldots\ldots a_9$. Then

(A) $\sum_{i=1}^{n} D_i = 1$ (B) $\sum_{i=1}^{n} D_i = 0$
(C) $D_i = D_j, \ \forall i, j$ (D) None of these

85. If A and B are square matrices of the same order and A is non singular, then for a positive integer $n, \left(A^{-1}BA\right)^n$ is equal to

(A) $A^{-n}B^nA^n$ (B) $A^nB^nA^{-n}$
(C) $A^{-1}B^nA$ (D) $n\left(A^{-1}BA\right)$

Space for Rough work

86. If both roots of equation $ax^2 + x + c - a = 0$ are imaginary and $c > -1$, then
(A) $3a > 2 + 4c$ (B) $3a < 2 + 4c$
(C) $c < a$ (D) none of these

87. The value of a for which the equation $x^3 + 2ax + 2 = 0$ and $x^4 + 2ax^2 + 1 = 0$ have a common root is
(A) $\frac{17}{8}$ (B) $-\frac{17}{8}$
(C) $\frac{15}{8}$ (D) $-\frac{15}{8}$

88. Let S be the set of 6-digit numbers $a_1a_2a_3a_4a_5a_6$ (all digits distinct) where $a_1 > a_2 > a_3 > a_4 < a_5 < a_6$. Then n(S) is equal to
(A) 210 (B) 2100
(C) 4200 (D) 420

89. 6 boys and 4 girls sit in a line so that p is the number of arrangements when no two girls sit together and q is the number of arrangements when all the girls sit together, then $\frac{p}{q}$ is
(A) 1 (B) $\frac{7}{2}$
(C) 5 (D) none of these

90. In a n sided regular polygon the probability that the two diagonal chosen at random will intersect inside the polygon is
(A) $\frac{2\,{}^nC_2}{({}^nC_2 - n)C_2}$ (B) $\frac{{}^{n(n-1)}C_2}{({}^nC_2 - n)C_2}$
(C) $\frac{{}^nC_4}{({}^nC_2 - n)C_2}$ (D) none of these

Space for Rough work